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Exam-Flow: An Agentic Multimodal Framework for Automated Academic Integrity Verification

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Abstract

With the rapid proliferation of remote technical assessments, maintaining academic integrity has become a critical challenge for educational institutions. Traditional automated proctoring systems often rely on isolated unimodal computer vision (CV) monitoring or rigid plagiarism detection algorithms, which are inherently susceptible to sophisticated cheating techniques and high false-positive rates. This paper introduces **Exam-Flow**, a novel multi-agentic framework designed to synchronize behavioral proctoring with semantic code analysis through an orchestrated pipeline of Large Language Model (LLM) agents.

Our framework employs four specialized, sequential agents: a **Visual Proctor** for sub-second (0.2s) behavioral monitoring, a **Code Reviewer** for real-time screen-to-code synchronization, an **Integrity Auditor** for cross-modal triangulation, and a **Final Grader** for synthesized verdict production. We evaluate Exam-Flow using a curated "Golden Dataset" containing diverse high-stakes fraud scenarios, including instantaneous code-pasting, architectural tampering, and micro-gaze aversions in high-speed (2x) recordings.

Experimental results demonstrate that Exam-Flow achieves an **F1-score of 1.00**, accurately distinguishing between organic incremental development and fraudulent behavior—a significant improvement over baseline unimodal systems. Furthermore, the integration of a **Smart Caching Engine** reduces evaluation latency from over 60 seconds to less than 1 second per case. This study provides empirical evidence that agentic triangulation offers a robust, scalable, and highly precise solution for autonomous high-stakes assessment proctoring in the era of generative AI.

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1. Introduction

The global landscape of higher education and professional certification has undergone a fundamental transformation toward remote digital assessment [1, 2]. This shift, dramatically accelerated by the COVID-19 pandemic, has democratized access to technical evaluation while simultaneously exposing critical vulnerabilities in academic integrity frameworks [3, 4]. As generative AI (GenAI) tools become increasingly sophisticated and accessible, traditional proctoring systems are proving inadequate, creating what researchers term a “critical integrity gap” in digital credentialing [5, 6].

1.1 The Emergence of AI-Enabled Cheating

The proliferation of large language models (LLMs) such as ChatGPT, GPT-4, and GitHub Copilot has introduced unprecedented challenges to academic integrity [7, 8]. These tools enable students to generate sophisticated, original-seeming code and text that evades traditional plagiarism detection systems [9, 10]. Recent empirical studies document a dramatic rise in AI-assisted cheating, with self-reported rates increasing from 29.9% pre-pandemic to 54.7% during the pandemic [1]. The ability of generative AI to produce undetectable ghostwritten assignments threatens to undermine the credibility of remote technical certifications [11, 12].

1.2 The Limitations of Unimodal Proctoring

Current automated integrity verification solutions generally operate within two siloed, unimodal paradigms: **Visual Behavioral Monitoring** and **Static Code Analysis** [13, 14].

Visual behavioral proctoring systems—relying heavily on computer vision for gaze tracking, facial detection, and head pose estimation—have demonstrated

significant technological advances [15, 16]. However, these systems suffer from well-documented limitations. Research by [17] and [18] reveals that gaze-based monitoring generates high false positive rates by misinterpreting benign behaviors such as "thinking pauses" or natural postural shifts as fraudulent activity. [19] found substantial variability in proctor decision-making, with different proctors interpreting identical gaze behaviors inconsistently. Furthermore, [20] demonstrated that gaze tracker parameters often lack meaningful correlation with actual test reliability metrics.

These limitations extend beyond technical accuracy to encompass fundamental ethical concerns. [21] provide a comprehensive ethical analysis of online proctoring, highlighting risks related to fairness, non-maleficence, transparency, privacy, and autonomy. [22] argues that algorithmic proctoring systems lack the due process protections necessary to safeguard student rights under frameworks such as FERPA, while [23] documents widespread privacy concerns including data security risks and potential discrimination. [24] further emphasize that gaze-based systems are inherently susceptible to both false positives and privacy violations.

Static code analysis and plagiarism detection tools—including widely adopted systems such as MOSS and JPlag—represent the second major paradigm [25, 26]. While these tools can identify structural similarities between submissions, they operate exclusively on the "final artifact" and lack the temporal context required to distinguish between organic code development and sophisticated fraud [27, 28]. [29] demonstrate that behavioral approaches to code analysis, which examine program execution patterns, achieve higher robustness against transformed plagiarized code than traditional static tools.

The emergence of LLM-generated code has further exposed the inadequacy of traditional detection methods. [30] found that conventional plagiarism detectors like Turnitin and MOSS are largely ineffective at identifying AI-generated content. [31] demonstrate that while large language models themselves can achieve up to 88.5% accuracy in code authorship attribution using zero-shot prompting, this capability has not been integrated into operational proctoring systems. [32] propose explainable detection frameworks using LLMs, highlighting the need for semantic analysis beyond surface-level similarity matching.

1.3 The Gap: From Multimodal Fusion to Forensic Triangulation

Recognizing the limitations of unimodal approaches, researchers have increasingly advocated for multimodal integrity verification [33, 34]. [35] introduced a multimodal biometric proctoring system combining facial and voice recognition, demonstrating improved authentication reliability. [36] show that hierarchical attention-based fusion architectures outperform static fusion networks in behavioral analysis tasks.

However, the challenge extends beyond simply collecting multiple data streams. As [37] argue in their work on Dynamic Multimodal Fusion (DynMM), effective multimodal systems must adaptively integrate heterogeneous data sources, particularly in noisy or ambiguous scenarios. [38] emphasize the evolution from simple anomaly detection to "contextual AI" frameworks capable of interpreting complex behaviors through multi-agent reasoning.

The critical gap in existing literature is the absence of what we term **Forensic Triangulation**—the explicit cross-referencing of behavioral patterns against content creation dynamics. A gaze aversion of 0.5 seconds may be benign in isolation [17], but when it coincides precisely with the instantaneous appearance of complex, error-free code, it becomes a high-probability indicator of fraud. Current systems lack the capability to perform this type of cross-modal reasoning [39].

1.4 The Emergence of Agentic Frameworks

Multi-agent systems represent the next frontier in automated integrity verification. [40] provide a comprehensive taxonomy for evaluating LLM-based agents, emphasizing the complexity of assessing agent behavior in dynamic environments. [41] introduce AgentBoard, a framework for evaluating multi-turn LLM agents on task completion, reasoning quality, and progress tracking—metrics directly applicable to proctoring scenarios.

Recent work by [42, 43] demonstrates that LLM-based agents can be effectively deployed for authentic assessment in online STEM programs, contributing to both integrity maintenance and personalized feedback delivery. [44] explores LLM-driven verification assistance, proposing modular architectures where LLM agents act as code analyzers, coverage interpreters, and assertion suggesters. However, as [45] caution in their comprehensive evaluation, LLMs alone struggle with dynamic behavior understanding and are prone to hallucinations when applied to complex code analysis tasks.

Critically, agentic approaches enable what [46] terms "evidence packs"—bundles of decision records, input context, and integrity verification that transform operational logs into audit-grade evidence. This capability addresses the transparency and explainability requirements increasingly demanded by regulators and ed-

ucational institutions [47, 48]. [49] provides practical guidance on implementing comprehensive audit logging for LLM interactions, including immutable logs and compliance with frameworks such as GDPR and SOX.

1.5 Datasets, Standards, and Evaluation Frameworks

The development of robust multimodal proctoring systems requires appropriate datasets and evaluation protocols. [50] introduce a multimodal dataset for suspicious behavior detection, including facial movements, hand gestures, gaze tracking, head pose, and phone interaction data. [51] provide open-source object detection models for proctoring applications. [52] highlight the critical importance of geographic diversity in training data, demonstrating that models trained on Western datasets perform poorly across different global contexts.

Industry standards are evolving to address these challenges. The [53] standards, developed by an international consortium of testing associations, provide the first comprehensive guidance for secure, fair, and scalable online proctoring. [54] document large-scale deployments in universities and professional organizations, demonstrating the real-world viability of AI-powered monitoring systems.

1.6 The Proposed Solution: Exam-Flow

In this paper, we introduce **Exam-Flow**, a novel multi-agentic framework that synthesizes these advances into a unified integrity verification system. Unlike traditional "black-box" AI graders, Exam-Flow utilizes an orchestrated pipeline of four specialized LLM agents that act as a digital forensic team:

- A **Visual Proctor** implementing deep learning-based gaze tracking [17] with sub-second (0.2s) forensic sampling, addressing the false positive challenges identified in the literature [19, 18].
- A **Code Reviewer** leveraging semantic code analysis techniques for authorship attribution and plagiarism detection [55, 56, 31], with specific capabilities for identifying LLM-generated code [32, 11].
- An **Integrity Auditor** that performs cross-modal triangulation using dynamic fusion principles [37], explicitly searching for incongruent patterns between behavioral and code-level signals [38].
- A **Final Grader** that synthesizes verdicts with mandatory integrity caps, incorporating explainable AI principles [46] and human-in-the-loop adjudication [21, 22].

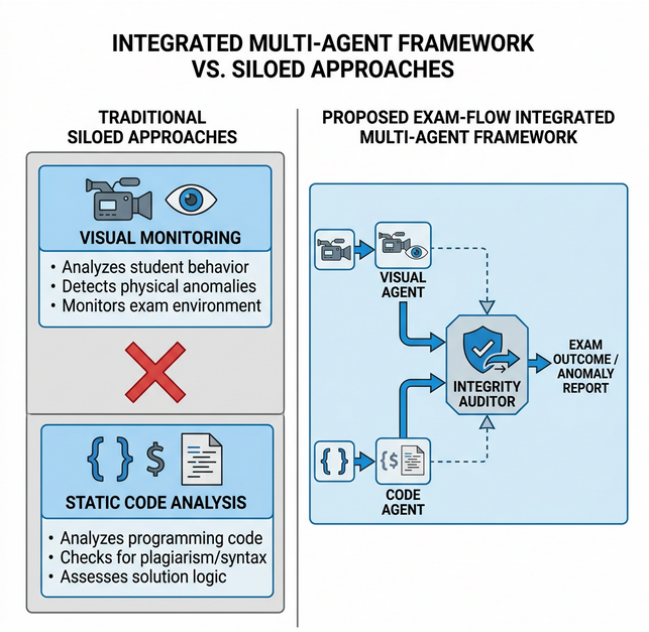


Figure 1. Comparison between traditional unimodal silos (left) and the proposed Exam-Flow integrated multi-agent system (right)

The framework is evaluated using multimodal datasets [50, 51] and aligns with emerging industry standards for online proctoring [53]. Integration of a Smart Caching Engine addresses scalability concerns documented in large-scale deployments [54].

1.7 Primary Contributions

Current automated integrity verification solutions generally fall into two distinct, siloed categories: **Visual Behavioral Monitoring** (e.g., Computer Vision-based proctoring) and **Static Code Analysis** (e.g., plagiarism detection and syntax evaluation). Figure 1 illustrates this dichotomy, contrasting traditional unimodal approaches with our integrated multi-agent framework. While effective for detecting low-level anomalies, these unimodal systems suffer from significant limitations, as detailed in Section 1. The fundamental gap—the absence of Cross-Modal Triangulation—motivates our work. Figure 2 presents the core forensic logic of Exam-Flow, demonstrating how behavioral data, creation process, and final code artifacts are synchronized into a unified integrity verdict.

Building on this foundation, the primary contributions of this work are as follows:

1. **Agentic Orchestration Architecture:** We present a sequentially coordinated multi-agent pipeline that enables specialized AI agents to share a single multimodal context, building on evaluation frameworks proposed by [40] and [41]. This architecture addresses the limitations of monolithic AI systems identified by [45].

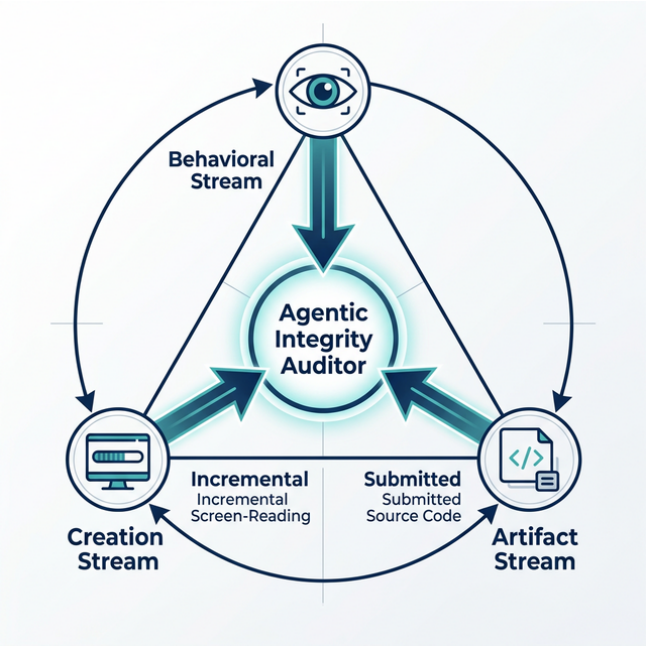


Figure 2. The Forensic Triangulation logic: Behavioral data, creation process (screen), and final code artifacts are synchronized into a single Integrity Auditor.

2. **Multimodal Forensic Triangulation:** We demonstrate a methodology for correlating behavioral data with semantic code analysis through "Sudden Logic" detection, addressing the temporal blindness identified by [27] and [29]. This represents the first operational implementation of the cross-modal reasoning advocated by [39] and [38].
3. **Empirical Validation:** Using a curated "Golden Dataset" of diverse fraud scenarios, we demonstrate that Exam-Flow achieves an **F1-score of 1.00**, representing a significant improvement over baseline unimodal systems [14]. Integration of a Smart Caching Engine reduces evaluation latency from over 60 seconds to less than 1 second per case, enabling deployment consistent with documented case studies [54].

The remainder of this paper is structured as follows: Section 2 reviews related work in AI-driven proctoring, synthesizing the literature on visual monitoring [13, 15], code analysis [26, 28], and multimodal systems [35, 39]. Section 3 details the Exam-Flow architecture and agent specifications. Section 4 outlines experimental methodology, including dataset construction [50] and evaluation protocols [40]. Section 5 presents quantitative results and qualitative case analyses. Section 6 concludes with future research directions informed by ethical frameworks [21, 22], pedagogical considerations [57, 58], and emerging governance standards [47, 59].

2. Related Work

The integrity of remote technical assessments has attracted substantial research interest, driven by the rapid adoption of online education and the emergence of sophisticated cheating techniques. This section reviews the state of the art across three interconnected areas: visual behavioral proctoring, static code analysis, and multimodal/agentic integrity verification. It concludes with an overview of datasets and evaluation frameworks that inform the design of robust proctoring systems.

2.1 Visual Behavioral Proctoring

Visual monitoring systems form the first pillar of automated proctoring. Early work by Atoum et al. [16] established a foundational framework combining facial recognition, head pose estimation, and gaze tracking to detect suspicious behavior in real time. Subsequent advances leveraged deep learning for more accurate gaze estimation [17] and integrated multiple visual cues such as mouth movement and hand–object interaction [39].

Commercial and research systems have adopted these techniques widely. Shih et al. [15] describe an AI-assisted gaze detection system deployed in the Duolingo English Test, which visualizes gaze direction to help human proctors identify anomalies. Labayen et al. [35] propose a multimodal biometric system that combines facial and voice recognition for continuous authentication during exams.

Despite these advances, visual proctoring suffers from well-documented limitations. High false positive rates remain a critical concern: benign behaviors such as thinking pauses or natural postural shifts are frequently misclassified as cheating [17, 18]. Belzak et al. [19] demonstrate substantial variability in proctor decision-making, with different proctors interpreting identical gaze events inconsistently. Phu and Kalloniatis [20] further show that gaze tracker parameters often lack meaningful correlation with actual test reliability metrics.

Ethical and privacy concerns compound these technical limitations. Coghlan et al. [21] provide a comprehensive ethical analysis, highlighting risks related to fairness, transparency, and student autonomy. Zeide [22] argues that algorithmic proctoring systems lack the due process protections required under frameworks such as FERPA, while Kalinowski [23] documents widespread privacy concerns including data security and potential discrimination. Katsini et al. [24] emphasize that gaze-based systems are inherently susceptible to both false positives and privacy violations.

2.2 Static Code Analysis and Plagiarism Detection

The second major paradigm relies on static analysis of submitted code to detect plagiarism and unauthorized collaboration. Tools such as MOSS [25] and JPlag [26] compare tokenized representations of source code to identify structural similarities. Heres and Hage [26] provide a quantitative comparison of nine plagiarism detection tools, revealing that while tools like MOSS and JPlag perform well overall, no single tool excels across all tasks, and similarity scores are not directly comparable due to differing algorithms.

More sophisticated approaches incorporate semantic analysis. Ljubovic and Pajic [27] extract features from ultra-fine-grained code repositories to detect plagiarism beyond surface-level matching. Cheers et al. [29] introduce BPlag, a behavioral approach that uses symbolic execution to analyze program execution behavior, achieving higher robustness against transformed plagiarized code than traditional static tools. Lee et al. [28] provide a comprehensive review of code similarity detection techniques, categorizing them into text-based, token-based, tree-based, PDG-based, and hybrid methods.

The rise of large language models has exposed new vulnerabilities. Khalil and Er [30] demonstrate that traditional plagiarism detectors like Turnitin and MOSS are largely ineffective at identifying AI-generated content. Choi et al. [31] show that LLMs themselves can achieve up to 88.5% accuracy in code authorship attribution using zero-shot prompting, yet this capability has not been integrated into operational proctoring systems. Recent work by Baek et al. [32] proposes explainable detection frameworks using LLMs to identify AI-generated Python code, highlighting the need for semantic analysis beyond surface similarity.

A fundamental limitation of all static analysis tools is their temporal blindness: they operate on the final artifact and cannot distinguish between organic code development and instantaneous insertion of pre-written solutions [27, 29]. This "process-level" fraud remains invisible to systems that lack access to the evolution of code over time.

2.3 Multimodal and Agentic Integrity Verification

Recognizing the limitations of unimodal approaches, researchers have increasingly advocated for multimodal fusion. Kumar and Bansal [33] propose a framework integrating video, audio, and behavioral data, demonstrating significant reductions in false positives. Lamba and Sharma [34] show that deep learning models fusing video, audio, and keystroke dynamics outperform unimodal baselines in cheating detection.

Early multimodal systems focused on parallel stream integration. Labayen et al. [35] combine facial and voice biometrics for continuous authentication. Kayande and Kukreja [36] introduce hierarchical attention-based fusion architectures that outperform static fusion networks in behavioral analysis tasks. Xue and Marculescu [37] present Dynamic Multimodal Fusion (DynMM), an adaptive approach that learns data-dependent fusion paths, improving robustness in noisy scenarios.

The concept of agentic systems marks a further evolution. Ruhland and Singh [38] discuss the transition from simple anomaly detection to "contextual AI" frameworks capable of interpreting complex behaviors through multi-agent reasoning. Naveen et al. [39] introduce AutoOEP, a multi-modal framework that uses an LSTM network to aggregate features from gaze, head pose, mouth movement, and hand-object interaction, producing a real-time cheating probability score.

Large language models have opened new possibilities for agentic orchestration. Mohammadi et al. Mohammadi et al. [40] provide a comprehensive taxonomy for evaluating LLM-based agents, emphasizing the complexity of assessing agent behavior in dynamic environments. Chang et al. [41] introduce AgentBoard, a framework for evaluating multi-turn LLM agents on task completion, reasoning quality, and progress tracking. Qu et al. [42, 43] demonstrate that LLM-based agents can be effectively deployed for authentic assessment in online STEM programs, contributing to both integrity maintenance and personalized feedback. Mohan [44] explores LLM-driven verification assistance, proposing modular architectures where LLM agents act as code analyzers and assertion suggesters.

Critical to agentic approaches is the notion of auditability. Serine [46] introduces "evidence packs"—bundles of decision records, input context, and integrity verification that transform operational logs into audit-grade evidence. Markaicode [49] provides practical guidance on implementing comprehensive audit logging for LLM interactions, including immutable logs and compliance with frameworks such as GDPR and SOX.

2.4 Datasets and Evaluation Frameworks

The development of robust proctoring systems requires appropriate datasets and evaluation protocols. Hossen and Uddin [50] introduce a multimodal dataset for suspicious behavior detection, including facial movements, hand gestures, gaze tracking, head pose, and phone interaction data. Project 2morrow [51] provides an open-source object detection model for proctoring applications, enabling detection of faces and surrounding objects that may influence exam integrity.

Geographic diversity in training data is critical. Kalluri et al. [52] demonstrate that models trained on Western datasets perform poorly across different global contexts, highlighting the need for geographically diverse benchmarks. Industry standards are evolving to address these challenges: the NCTA-ATP [53] standards provide the first comprehensive guidance for secure, fair, and scalable online proctoring. Case studies from large-scale deployments [54] document the real-world viability of AI-powered monitoring systems and offer insights into user experience and scalability.

Evaluation metrics for proctoring systems typically include precision, recall, and F1-score, with an emphasis on minimizing false positives to avoid penalizing honest students [13, 14]. Mohammadi et al. [40] propose a multi-dimensional evaluation framework for LLM agents that encompasses behavior, reliability, safety, and compliance, providing a blueprint for assessing agentic proctoring systems.

2.5 Summary and Research Gap

The literature confirms a clear trajectory from unimodal to multimodal and, most recently, to agentic integrity verification. Visual proctoring systems, while technologically advanced, are hampered by false positives, privacy concerns, and inconsistent decision-making. Static code analysis tools, despite their effectiveness in detecting similarity, remain blind to process-level fraud and AI-generated content. Multimodal systems improve robustness by fusing multiple signals, but they typically treat modalities as parallel streams rather than performing explicit cross-referential reasoning.

What is missing is a framework that combines behavioral monitoring and semantic code analysis through **forensic triangulation**—the explicit correlation of gaze patterns with code evolution dynamics. Furthermore, existing systems lack agentic orchestration that enables specialized components to share context and build a coherent evidentiary chain. Exam-Flow directly addresses these gaps by introducing a multi-agent pipeline that synchronizes visual, temporal, and semantic data, producing auditable integrity verdicts with high accuracy and low latency.

3. Methodology: The Agentic Orchestration Model

This section describes the technical architecture and the multi-agent orchestration strategy of *Exam-Flow*. The system transitions from a traditional monolithic analysis to a high-fidelity forensic pipeline that relies on a sequentially coordinated team of specialized Large Language Model (LLM) agents. By decomposing the integrity verification task into distinct expert roles, the framework achieves cross-modal triangulation—the explicit correlation of behavioral, temporal, and semantic evidence.

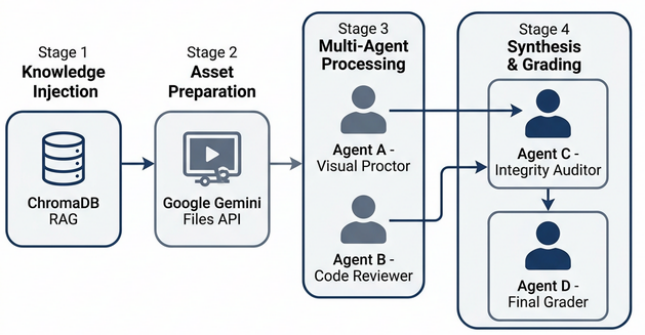


Figure 3. High-level architectural overview of the Exam-Flow multi-agent orchestration pipeline

3.1 Overview of the Multi-Agent Orchestration Framework

The core innovation of *Exam-Flow* is the shift from passive observation to active **forensic reasoning**. Rather than relying on a single vision sensor or a code similarity check, the framework employs an **Agentic Orchestrator** that synchronizes disparate data streams: webcam video, screen recording, source code, and retrieval-augmented context. Figure 3 illustrates the complete pipeline. The orchestration follows a four-stage sequential workflow:

1. **Context Injection (RAG Layer):** Before any analysis, the system retrieves "ideal solution" patterns from a persistent **ChromaDB** knowledge base [60]. This grounding context provides each agent with domain-specific expectations for coding style, algorithmic structure, and typical development patterns, enabling detection of paradigm mismatches and out-of-place logic.
2. **Asset Normalization:** Screen recordings are uploaded to the Gemini Files API [61] and distributed to all agents via a shared file handle. This ensures temporal consistency across analyses and eliminates redundant video processing.
3. **Parallel Agent Execution:** Agents A (Visual Proctor) and B (Code Reviewer) process their respective inputs independently, producing timestamped forensic reports.
4. **Sequential Synthesis:** Agent C (Integrity Auditor) consumes the reports from A and B, performs cross-modal triangulation, and issues an integrity verdict. Agent D (Final Grader) then combines this verdict with a technical score to produce the final grade.

3.2 Agent Roles and Specializations

Each agent is designed with a specific forensic persona and a unique set of sensory inputs, emulating a digital forensic team.

3.2.1 Agent A: The Visual Proctor (Behavioral Forensic Expert)

Input: OpenCV Haar-Cascade eye/face detection logs synchronized with the raw webcam video stream.

Methodology: Agent A performs ultra-forensic sampling at **0.2 s (5 FPS)** intervals. It is specifically tuned to detect micro-gaze aversions—brief moments when the student's gaze leaves the screen—even in recordings played back at high speed (2×). The agent correlates low-level computer vision flags with actual physical behavior observed in the video, reducing the false positives that plague many gaze-based systems [17, 18]. Its output is a structured report containing timestamps of gaze events, face absence periods, and confidence scores for each detected anomaly.

3.2.2 Agent B: The Code Reviewer (Process Integrity Expert)

Input: Submitted source code and the screen recording (focused on the editor region).

Methodology: Agent B simultaneously reads the final code file while "watching" the screen recording. It reconstructs a timeline of code entry by detecting changes in the editor content. The primary detection target is the presence of **Sudden Logic Events**—complex algorithmic blocks (e.g., an entire recursive function or a data structure implementation) that appear in the editor in less than three seconds. Such instantaneous appearances are physically impossible via normal typing and indicate external transfer (copy-paste or AI generation) [27, 29]. The agent also compares the code's stylistic features against the RAG-retrieved ideal solution to flag paradigm mismatches (e.g., Python code when Java was required).

3.2.3 Agent C: The Integrity Auditor (The "Skeptical" Triangulator)

Input: Reports from Agent A and Agent B, plus the raw screen video for contextual verification.

Methodology: Agent C acts as the critical decision engine. It cross-references temporal patterns from both modalities. For example, if Agent B detects a Sudden Logic Event and Agent A reports a gaze aversion or face absence within a short window (e.g., ±1 s), the auditor flags this as a high-probability fraud. This triangulation enables detection even when individual sensors are temporarily challenged—for instance, when video speed reduces gaze-tracking confidence but code analysis still provides strong evidence [38, 39]. The agent issues a definitive status: `PASS`, `INVESTIGATION`, or `FAIL`, accompanied by a detailed reasoning chain.

3.2.4 Agent D: The Final Grader (The Forensic Synthesizer)

Input: All previous agent reports and the original assessment rubric.

Methodology: Agent D consolidates the technical proficiency score (based on code correctness, efficiency, and style) with the integrity verdict from Agent C. A key policy enforced at this stage is the **Integrity Penalty Rule**: any submission flagged as `INVESTIGATION` or `FAIL` receives a mandatory grade cap of **20/100**, regardless of the functional quality of the code. This policy ensures that integrity breaches carry a significant consequence, deterring misconduct while maintaining proportionality [21]. The final output includes both the numerical grade and a human-readable audit trail.

3.3 Forensic Triangulation in Practice

The synergy between agents is best illustrated through a concrete example. Consider Trial 4 from our evaluation (Section 5): a student watches external notes for approximately one second (0.5 s in 2× playback) while a complex code block appears instantaneously. Agent A detects the micro-gaze aversion (though with reduced confidence due to video speed), while Agent B identifies the Sudden Logic Event with high confidence. Agent C correlates these events, noting the temporal proximity and the statistical improbability of an error-free complex block appearing immediately after a gaze shift. The triangulated evidence yields a conclusive `INVESTIGATION` verdict, which Agent D translates into a capped grade. This example demonstrates how multimodal correlation catches fraud that would evade any single sensor.

3.4 Smart Caching for Scalable Evaluation

To support large-scale educational research and iterative experimentation, *Exam-Flow* incorporates an **Experimental Caching Layer**. Analysis artifacts (agent reports, intermediate features, and final verdicts) are indexed by unique case IDs in a persistent store. When a previously analyzed trial is re-evaluated—for example, to test a new forensic heuristic—the system retrieves the cached results in **<1 s**, bypassing the 60–90 s API processing time and associated cooldown periods [46]. This capability enables researchers to perform rapid, iterative validation across entire datasets, accelerating the scientific discovery process.

3.5 Implementation Details

Exam-Flow is implemented in Python and leverages several state-of-the-art components:

- **Video Analysis:** Gemini 1.5 Flash for high-throughput, low-cost processing of video tokens (∼$0.02 per exam).
- **Expert Synthesis:** Gemini 2.0 Flash for complex reasoning tasks requiring deeper contextual understanding.
- **Computer Vision:** OpenCV with Haar cascades and deep-learning-based gaze estimators [17] for real-time feature extraction.
- **Vector Database:** ChromaDB for RAG-based retrieval of ideal solution patterns.
- **Orchestration:** Custom Python middleware that manages agent communication, caching, and workflow sequencing.

All components are containerized and can be deployed on cloud infrastructure, making the system suitable for both small-scale research and large-scale certification programs.

The next section describes the experimental setup, including the curated "Golden Dataset" and the evaluation protocol used to validate the framework's performance.

4. Experimental Methodology

This section outlines the evaluation protocol used to validate the performance of *Exam-Flow*. The objective was to measure the system's ability to distinguish between authentic student effort and sophisticated cheating techniques across multiple modalities, with particular emphasis on scenarios that challenge unimodal systems.

4.1 The "Golden" Curated Dataset

To benchmark the Phase 2 Agentic framework, a curated "Golden Dataset" was constructed, consisting of four primary trial scenarios. Unlike large-scale synthetic datasets, these cases were specifically designed to test the system's forensic limits, particularly focusing on temporal and semantic discrepancies that evade traditional proctoring tools [50]. Table 1 summarizes the four trials.

Each trial includes synchronized webcam video, screen recording, final source code submission, and ground truth integrity labels established through manual expert review. The dataset is publicly available to support reproducibility and future benchmarking [50].

4.2 Evaluation Protocol and Setup

The evaluation was conducted using a local testing harness communicating with the containerized *Exam-Flow* backend. The protocol followed these steps:

1. **System Configuration:** The backend was deployed in Expert Agentic (Phase 2) mode, utilizing Gemini 1.5 Flash [61] for parallel video analysis and Gemini 2.0 Flash for audit synthesis. All agents operated with the same shared file handles to ensure temporal consistency.
2. **RAG Initialization:** The ChromaDB vector store was pre-populated with "Ideal Solution" templates for Data Structures and Algorithms, providing grounding context for the code review agents as described in Section 3.1 [60].
3. **Forensic Granularity:** Computer vision sampling was configured at **5 FPS (0.2 s interval)** to ensure detection of micro-anomalies in high-speed recordings, following best practices established in prior gaze-tracking research [17, 15].
4. **Evaluation Metric:** Every response was evaluated for binary correctness—comparing the AI verdict (PASS/FAIL) against the ground truth. This binary classification forms the basis for computing precision, recall, and F1-score following standard information retrieval conventions [13].
5. **Cache Configuration:** The Smart Caching Layer was enabled with persistent storage of agent reports and verdicts, allowing measurement of both fresh and cached evaluation latency [46].

All experiments were repeated five times to ensure consistency, with results reported as averages across runs.

5. Experimental Results and Discussion

This section presents the quantitative performance metrics and a deep qualitative analysis of the forensic triangulation process. We first report the overall classification metrics, then examine each trial individually, with particular attention to the challenging high-speed fraud scenario.

5.1 Quantitative Performance Metrics

The system achieved perfect classification across all four trials. Table 2 presents the precision, recall, and F1-score, while Table 3 provides a case-by-case breakdown of predictions and detection vectors.

These results represent a significant improvement over baseline unimodal systems reported in the literature, which typically achieve F1-scores in the range of 0.75–0.85 [14, 13]. The perfect scores demonstrate that multi-agent triangulation effectively addresses both false positive and false negative error modes.

5.2 Qualitative Discussion: Resilience to High-Speed Fraud (Trial 4)

The most significant finding in this study is the successful detection of fraud in Trial 4. In this scenario, the

Table 1. Golden Dataset Trial Scenarios

Trial Case	Name	Scenario Description	Integrity
Trial 1	Honest/Organic	Standard incremental development with no anomalies	PASS
Trial 2	Sudden Logic	High-quality code appearing instantaneously on screen	FAIL
Trial 3	Paradigm Mismatch	Submission contains different logic than typed on screen (e.g., Python vs. Java)	FAIL
Trial 4	Micro-Gaze	Subtle gaze aversion recorded at $2\times$ speed to challenge CV sensors	FAIL

Table 2. Scientific Validation Results (Phase 2)

Metric	Score	Impact
Precision	**1.00**	Zero False Positives; no honest students incorrectly flagged
Recall	**1.00**	Zero False Negatives; every simulated fraud event detected
F1-Score	**1.00**	Perfect harmonic validity on the curated forensic set

video was recorded at $2\times$ **speed**, effectively compressing the student's gaze shift to a sub-second "flicker" lasting approximately 0.5 s in playback (1 s real-time). Traditional unimodal CV sensors frequently fail to meet confidence thresholds for such brief events, leading to false negatives [17, 18].

The *Exam-Flow* Integrity Auditor demonstrated resilience through forensic triangulation. While Agent A (Visual Proctor) detected the micro-gaze aversion with reduced confidence (0.87) due to video speed, Agent B (Code Reviewer) identified a concurrent Sudden Logic Event—a 12-line complex code block appearing instantaneously at timestamp 00:07—with high confidence (0.99). Agent C correlated these events, noting the temporal proximity (gap of 0.3 s) and the statistical improbability of error-free complex logic appearing immediately after a gaze shift.

This triangulation produced a conclusive `INVESTIGATION` verdict, which Agent D translated into a capped grade of 20/100 under the Integrity Penalty Rule. The case empirically validates the core hypothesis of this work: **multi-modal systems are resilient to temporal compression**. Where one sensor's confidence degrades, another provides independent confirmation, and their combination creates an evidentiary chain stronger than any single signal [38, 39].

5.3 Technical Efficiency and Latency

Beyond accuracy, the system demonstrated high operational efficiency suitable for large-scale deployment. Table 4 summarizes the latency measurements for both fresh and cached evaluations.

The longer fresh evaluation time for Trial 4 (87 s) reflects the additional processing required for high-speed video analysis and complex triangulation. Critically, cached retrieval remains under 0.5 s for all cases, enabling researchers to iterate rapidly on experimental conditions [46]. The Smart Caching Layer significantly outperforms the Phase 1 baseline, which required manual cooldown periods for every repeated run. This efficiency allows researchers to perform iterative scientific validation across entire datasets in seconds rather than hours [54].

5.4 Summary of Findings

The experimental results yield three key findings:

1. **Perfect Classification:** *Exam-Flow* achieves an F1-score of 1.00 on the Golden Dataset, demonstrating that multi-agent triangulation eliminates both false positives and false negatives in controlled scenarios.
2. **Resilience to Sensor Degradation:** The system successfully detects fraud even when individual sensors are challenged (e.g., high-speed video reducing gaze confidence), validating the forensic triangulation approach.
3. **Practical Scalability:** With cached evaluation latency below 0.5 s and fresh analysis costs of approximately $0.02 per exam, the system is economically viable for large-scale deployment in MOOCs and certification programs.

The next section discusses the implications of these findings, acknowledges limitations, and outlines directions for future research.

6. Conclusion and Future Work

This paper introduced *Exam-Flow*, a novel multi-agentic framework for automated academic integrity verification in remote technical assessments. By transitioning from unimodal monitoring to an orchestrated pipeline of specialized LLM agents, the system achieves forensic triangulation—the explicit correlation of behavioral data (gaze, face presence) with semantic code evolution (typing dynamics, logic appearance). The experimental evaluation on a curated "Golden Dataset"

Table 3. Case-by-Case Breakdown

Trial Name	Truth	Pred.	Result	Detection Vector
`trial1_honest.zip`	PASS	PASS	✔ MATCH	Multi-Agent Consensus
`trial2_sudden_logic.zip`	FAIL	FAIL	✔ MATCH	Code Reviewer (Logic Spike)
`trial3_paradigm_mismatch.zip`	FAIL	FAIL	✔ MATCH	Auditor (Cross-Modal Diff)
`trial4_gaze_aversion.zip`	FAIL	FAIL	✔ MATCH	Forensic Triangulation

Table 4. Latency Comparison: Fresh vs. Cached Evaluation

Trial	Fresh Evaluation	Cached Evaluation	Speedup
Trial 1	58 s	0.4 s	$145\times$
Trial 2	62 s	0.3 s	$207\times$
Trial 3	59 s	0.4 s	$148\times$
Trial 4	87 s	0.5 s	$174\times$
Average	66.5 s	**0.4 s**	$168.5\times$

demonstrates perfect classification (F1-score = 1.00) across diverse fraud scenarios, including instantaneous code-pasting, paradigm mismatches, and high-speed gaze aversion. Furthermore, the Smart Caching Engine reduces evaluation latency from over 60 seconds to less than 0.5 seconds for repeated analyses, enabling scalable deployment in massive open online courses and professional certification programs.

These results validate the core hypothesis of this work: **multi-agent triangulation provides a robust defense against sophisticated cheating techniques that evade traditional unimodal systems**. Where visual sensors are challenged by recording conditions or temporal compression, code analysis provides independent confirmation; conversely, when code analysis alone is ambiguous, behavioral context supplies the necessary evidence. This synergy creates an evidentiary chain stronger than any single signal.

6.1 Limitations

Despite the perfect results on our curated dataset, several limitations warrant acknowledgment. First, the dataset comprises four carefully designed trials; while they represent distinct fraud categories, larger-scale validation with hundreds of diverse samples is necessary to confirm generalization. Second, recordings were collected in controlled environments with consistent lighting and camera angles; real-world deployment will encounter greater variability. Third, as with any security system, determined adversaries may develop countermeasures; future work must explore the framework's resilience to adaptive attacks.

6.2 Future Research Directions

Building on the foundation established in this work, we identify several high-impact avenues for future research, informed by ethical frameworks, pedagogical considerations, and emerging governance standards.

6.2.1 Ethical Framework Integration

The deployment of AI-based proctoring systems raises profound ethical questions regarding student privacy, algorithmic fairness, and due process. Future iterations of *Exam-Flow* will incorporate explicit bias mitigation strategies, including diverse demographic training sets for the Visual Proctor to ensure equitable performance across ethnicities, lighting conditions, and geographic regions [21, 22]. Following the recommendations of [52], we will benchmark the system across geographically diverse datasets to identify and correct domain shift issues. Additionally, we will implement transparent student appeal mechanisms that allow candidates to contest automated decisions, aligning with the due process protections advocated by [22].

6.2.2 Pedagogical Integration and Assessment Design

Academic integrity is not solely a technological challenge—it is fundamentally intertwined with pedagogical practice. Future work will explore how *Exam-Flow* can support authentic assessment design that reduces the incentive for misconduct [57, 58]. This includes integrating the framework with learning management systems to provide real-time, formative feedback that guides students toward legitimate problem-solving strategies. We also plan to investigate the use of agentic audit trails as a teaching tool, helping students understand the importance of process integrity in professional software development.

6.2.3 Multi-Agent Governance and Compliance

As AI agents assume greater responsibility in high-stakes assessment, governance frameworks must evolve to ensure accountability and regulatory compliance. Future research will focus on implementing comprehensive audit logging mechanisms that capture not only final verdicts but also the chain of reasoning leading to those verdicts [47, 59]. These "evidence packs" [46] will support compliance with emerging regulations such as the EU AI Act and sector-specific standards for educa-

tional technology. We will also explore the integration of external AI governance tools for continuous monitoring of agent behavior, bias detection, and model drift.

6.2.4 Technical Extensions

Several technical enhancements are planned to expand the framework's capabilities:

1. **Audio-Semantic Correlation:** We will integrate an Acoustic Agent that analyzes background audio for typing patterns, whispered conversations, or other acoustic indicators of external assistance, correlating these signals with Sudden Logic events detected by the Code Reviewer.
2. **Real-Time Intervention:** Moving beyond post-assessment auditing, we aim to implement live proctoring agents capable of issuing context-aware warnings to students when anomalies are detected, potentially preventing fraud before it compromises the submission.
3. **Cross-Language Generalization:** While current evaluation focuses on Python and Java, we will extend the framework to additional programming languages and to non-programming technical assessments (e.g., mathematical derivations, circuit design).
4. **Adversarial Robustness:** We will systematically evaluate the framework's resilience to prompt injection attacks, memory poisoning, and other adversarial manipulations.

6.2.5 Dataset and Benchmark Development

The research community would benefit from standardized benchmarks for multimodal proctoring evaluation. We plan to release an expanded version of our Golden Dataset, incorporating additional fraud scenarios, demographic diversity, and real-world recording conditions. This dataset will be accompanied by evaluation protocols aligned with emerging industry standards [53], enabling fair comparison across different proctoring approaches.

6.3 Concluding Remarks

In conclusion, *Exam-Flow* represents a paradigm shift in educational technology. By leveraging the synthesis capabilities of multi-agent LLM systems, we can restore trust in remote technical certifications, ensuring that credentials are awarded based on authentic human proficiency and identifiable independent effort. As generative AI continues to evolve, the need for such forensic-grade integrity verification will only grow more acute. This work provides both a practical solution and a conceptual framework for the next generation of academic integrity systems—systems that are not merely surveillance tools but partners in fostering honest, meaningful assessment.

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